Divyam Ahuja

+91 7015600547 ahujadivyam@gmail.com divyam.dev in/ahujadivyam git.divyam.dev gh/ahujadivyam

Experience

Microsoft Noida
Software Engineer Mar 2025 - Present
Tech Stack: Python, TypeScript, Docker, Azure, React, Playwright

- Engineer and evaluate AI-powered experiences across multimodal models and agentic workflows for the Excel Copilot team.
- Owned the end-to-end Responsible AI (RAI) evaluation pipeline gating model releases; optimized execution latency by **50%** through asynchronous worker pools, intelligent request batching, and custom retry algorithms.
- Onboarded next-generation multimodal AI agents into the core evaluation platform, designing pluggable backend adapters, isolated container sandboxes (Docker), and concurrent distributed cloud runtimes.
- Designed and implemented a parallelized automation framework for synthetic data generation and evaluation; boosted throughput by **5x** and generated **70,000+** trajectories across **20,000+** enterprise workbooks.
- Served as on-call engineer driving infrastructure reliability across evaluation pipelines—triaging partner team issues, resolving live-site instabilities, and handling incident response to maintain **>95%** reliability.
- Integrated comprehensive telemetry pipelines and KQL aggregations processing **millions** of daily events, establishing single-pane system health dashboards and automated triage.

Advanced Micro Devices (AMD) Bangalore
System Software Designer Aug 2022 - Feb 2025
Tech Stack: C/C++, Java (OpenJDK), Linux perf, Python, Django, React, SQLite, Docker

- Engineered an OpenJDK performance monitoring and telemetry platform (Django, React, SQLite) to track nightly SPECjbb runs, enabling early detection and isolation of performance regressions.
- Automated systems to benchmark and tune Java nightly using machine learning models, achieving a **30%** reduction in manual effort and improved parameter accuracy.
- Applied advanced Java tuning techniques and profiled CPU/memory bottlenecks with Linux perf, yielding a **2%** throughput increase on SPECjbb benchmarks across High Performance Computing (HPC) servers.
- Refactored and containerized legacy benchmarking workflows with Docker for cross-platform execution, driving a **90%** reduction in deployment and environment errors.
- Awarded the **Spotlight Award** for exceptional contributions to automation, tuning, and nightly benchmark visualization.

Chromium Remote
Google Summer of Code Jun 2022 - Aug 2022
Tech Stack: C++, WebAssembly, WebAudio, WebWorkers

- Engineered a lock-free C++ ring buffer as a reference library for real-time audio streaming between WebAudio, WebAssembly, and WebWorkers, benchmarking multiple concurrency strategies.
- summerofcode.withgoogle.com

Skills

Programming Languages: C/C++, Python, Go (Golang), Java, Kotlin, TypeScript, Bash
Frameworks & Tools: PyTorch, Azure OpenAI, Android SDK / NDK, React/Next.js, Django/Flask, Node.js, gRPC, Emscripten, Playwright
Infrastructure & Databases: Kubernetes, Docker, Linux/POSIX, PostgreSQL, MongoDB, Redis, MySQL, SQLite, WebAssembly, WireGuard

Projects

Fastbin: A Code Sharing Webapp | HTMX, Go (Golang), gRPC, PostgreSQL, Docker, Kubernetes
- Built a high-performance code sharing backend in Go utilizing gRPC Protobuf microservices, PostgreSQL with connection pooling and optimized indexing, containerized with Docker and deployed on Kubernetes.
- git.divyam.dev/divyam/fastbin, fastbin.lab.divyam.dev

Homelab | Raspberry Pi 5, WireGuard, Kubernetes, Rancher, MetalLB
- Personal infrastructure hosting containerized services with WireGuard VPN tunneling, MetalLB load balancing, and Rancher-managed Kubernetes, enabling secure, high-availability external access.

Education

Deenbandhu Chhotu Ram University of Science and Technology Haryana, India
Bachelor of Technology (B.Tech.) Computer Science Aug 2018 - Aug 2022